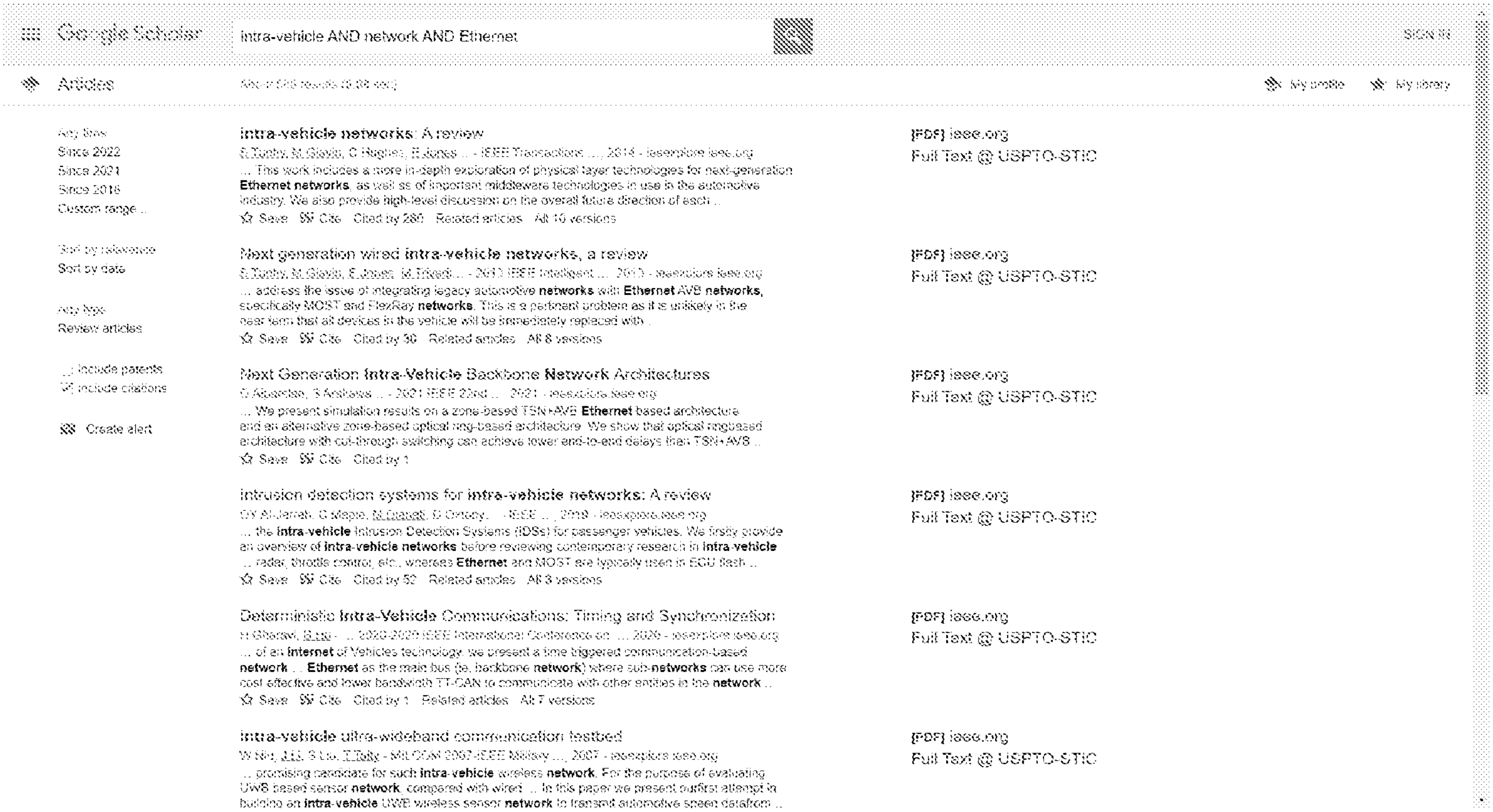

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… address the issue of integrating legacy automotive **networks** with **Ethernet** AVB **networks**, specifically MOST and FlexRay **networks**. This is a pertinent problem as it is unlikely in the near term that all devices in the vehicle will be immediately replaced with …
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… We present simulation results on a zone-based TSN+AVB **Ethernet** based architecture and an alternative zone-based optical ring-based architecture. We show that optical ringbased architecture with cut-through switching can achieve lower end-to-end delays than TSN+AVB …
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… the **intra-vehicle** Intrusion Detection Systems (IDSs) for passenger vehicles. We firstly provide an overview of **intra-vehicle networks** before reviewing contemporary research in **intra-vehicle** … radar, throttle control, etc., whereas **Ethernet** and MOST are typically used in ECU flash …
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… of an **Internet** of Vehicles technology, we present a time triggered communication-based **network** … **Ethernet** as the main bus (ie, backbone **network**) where sub-**networks** can use more cost effective and lower bandwidth TT-CAN to communicate with other entities in the **network** …
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… promising candidate for such **intra-vehicle** wireless **network**. For the purpose of evaluating UWB based sensor **network**, compared with wired … In this paper we present ourfirst attempt in building an **intra-vehicle** UWB wireless sensor **network** to transmit automotive speed datafrom …
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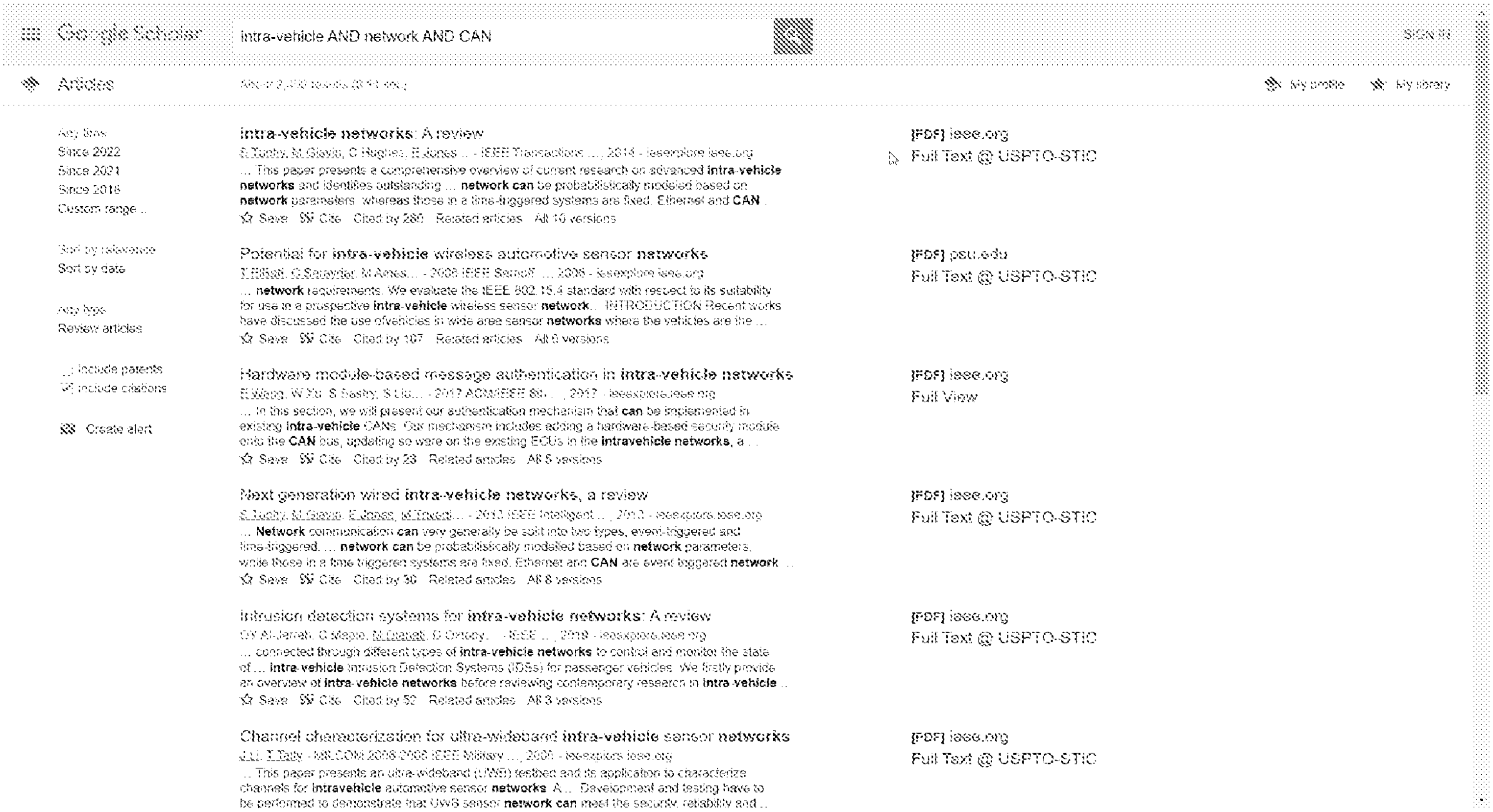

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… This paper presents a comprehensive overview of current research on advanced **intra-vehicle networks** and identifies outstanding … **network can** be probabilistically modeled based on **network** parameters, whereas those in a time-triggered systems are fixed. Ethernet and **CAN** …
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… **network** requirements. We evaluate the IEEE 802.15.4 standard with respect to its suitability for use in a prospective **intra-vehicle** wireless sensor **network**. … INTRODUCTION Recent works have discussed the use ofvehicles in wide area sensor **networks** where the vehicles are the …
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E Wang, W Xu, S Sastry, S Liu… - 2017 ACM/IEEE 8th …, 2017 - ieeexplore.ieee.org
… In this section, we will present our authentication mechanism that **can** be implemented in existing **intra-vehicle** CANs. Our mechanism includes adding a hardware-based security module onto the **CAN** bus, updating so ware on the existing ECUs in the **intravehicle networks**, a …
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… **Network** communication **can** very generally be split into two types, event-triggered and time-triggered. … **network can** be probabilistically modelled based on **network** parameters, while those in a time triggered systems are fixed. Ethernet and **CAN** are event triggered **network** …
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… connected through different types of **intra-vehicle networks** to control and monitor the state of … **intra-vehicle** Intrusion Detection Systems (IDSs) for passenger vehicles. We firstly provide an overview of **intra-vehicle networks** before reviewing contemporary research in **intra-vehicle** …
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… This paper presents an ultra-wideband (UWB) testbed and its application to characterize channels for **intravehicle** automotive sensor **networks**. A … Development and testing have to be performed to demonstrate that UWB sensor **network can** meet the security, reliability and …
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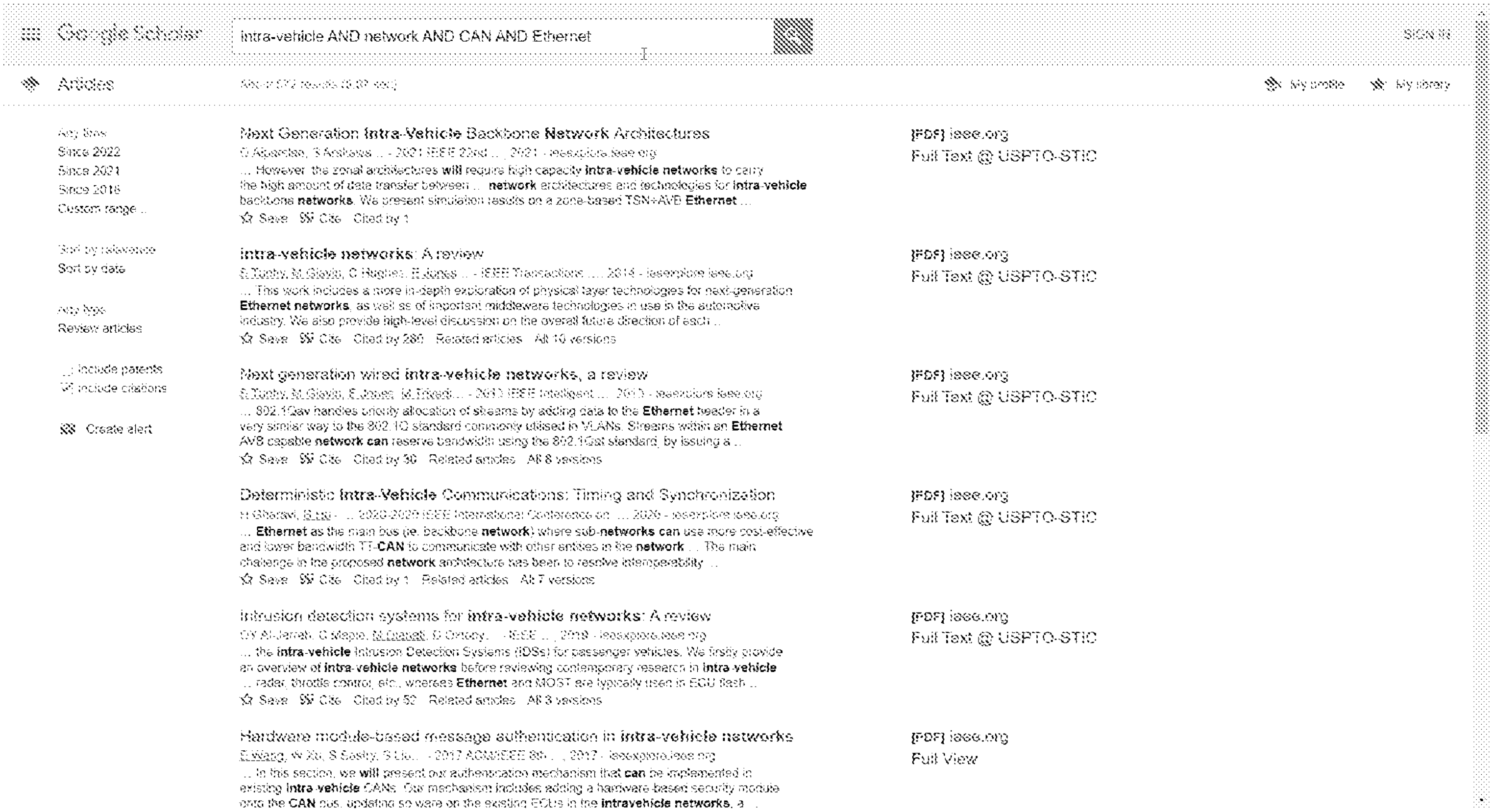

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... 802.1Qav handles priority allocation of streams by adding data to the **Ethernet** header in a very similar way to the 802.1Q standard commonly utilised in VLANs. Streams within an **Ethernet** AVB capable **network can** reserve bandwidth using the 802.1Qat standard, by issuing a ...
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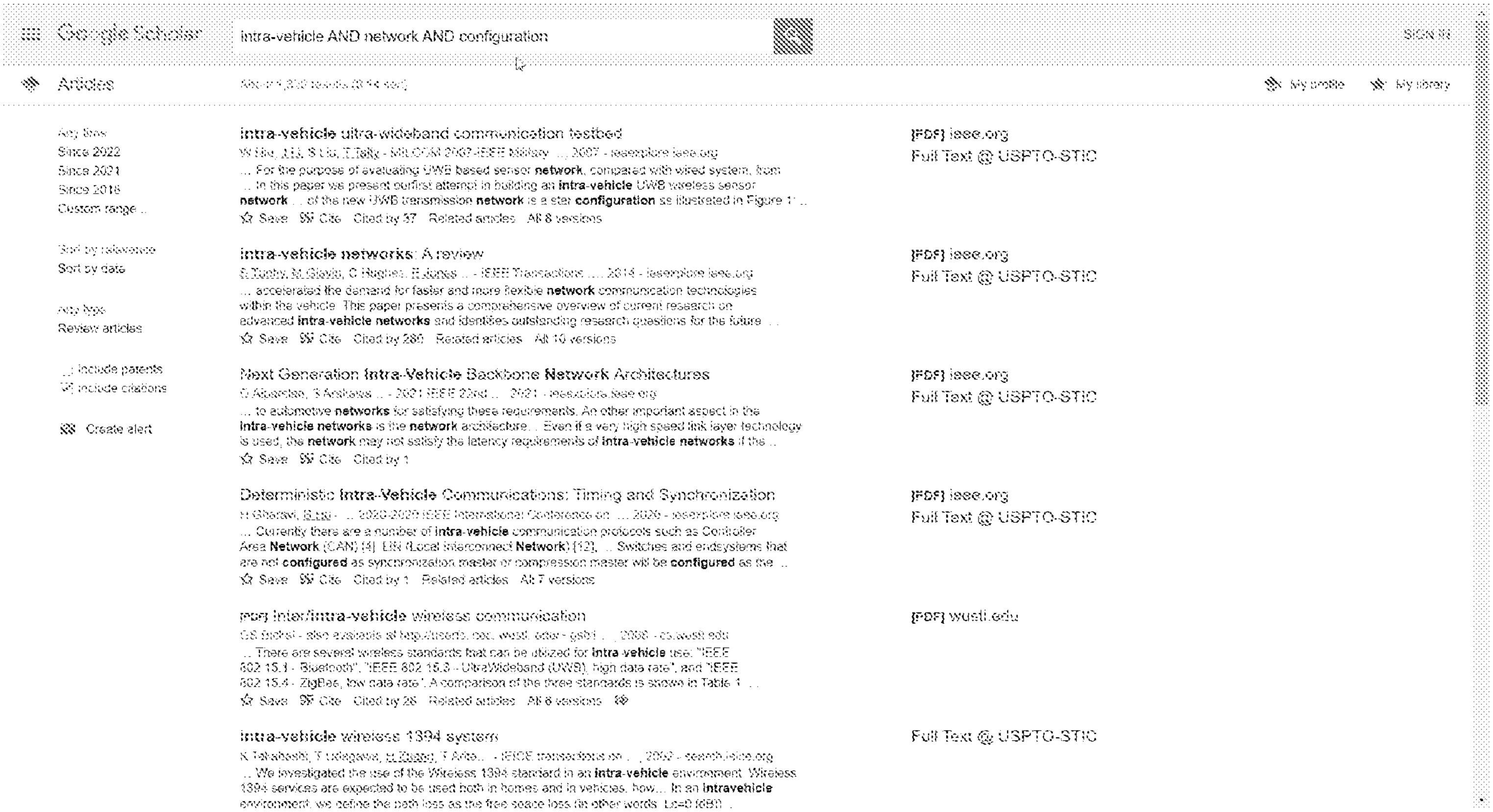

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… For the purpose of evaluating UWB based sensor **network**, compared with wired system, from … In this paper we present ourfirst attempt in building an **intra-vehicle** UWB wireless sensor **network** … of the new UWB transmission **network** is a star **configuration** as illustrated in Figure 1 …

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… to automotive **networks** for satisfying these requirements. An other important aspect in the **intra-vehicle networks** is the **network** architecture. … Even if a very high speed link layer technology is used, the **network** may not satisfy the latency requirements of **intra-vehicle networks** if the …

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… Currently there are a number of **intra-vehicle** communication protocols such as Controller Area **Network** (CAN) [4], LIN (Local Interconnect **Network**) [12], … Switches and endsystems that are not **configured** as synchronization master or compression master will be **configured** as the …

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… There are several wireless standards that can be utilized for **intra-vehicle** use: "IEEE 802.15.1 - Bluetooth", "IEEE 802.15.3 - UltraWideband (UWB), high data rate", and "IEEE 802.15.4 - ZigBee, low data rate". A comparison of the three standards is shown in Table 1. …

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K Takahashi, T Udagawa, H Zhang, T Arita… - IEICE transactions on …, 2002 - search.ieice.org

… We investigated the use of the Wireless 1394 standard in an **intra-vehicle** environment. Wireless 1394 services are expected to be used both in homes and in vehicles, how… In an **intravehicle** environment, we define the path loss as the free space loss (in other words, Lc=0 (dB)) …

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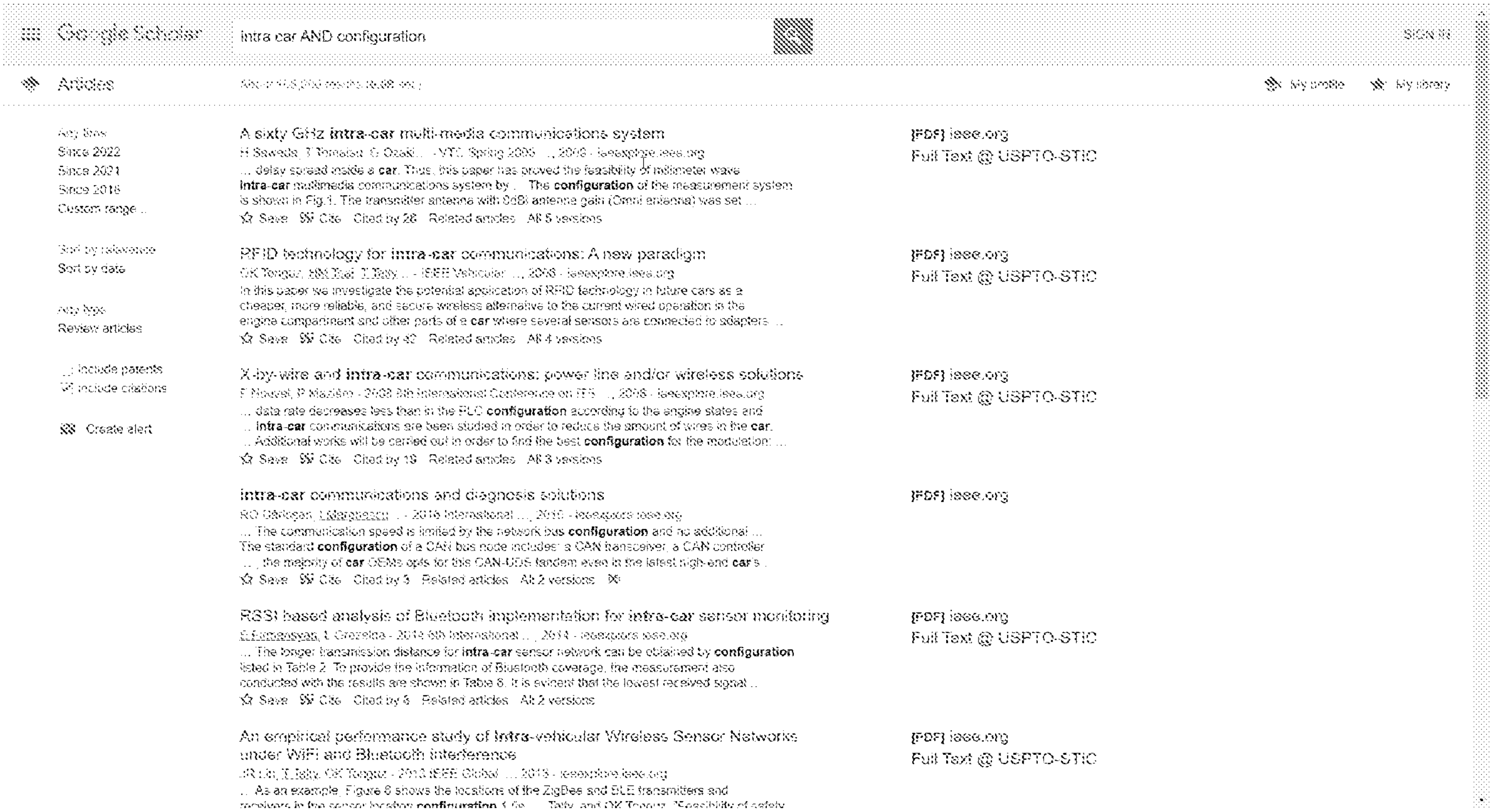

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... data rate decreases less than in the PLC **configuration** according to the engine states and ... **intra-car** communications are been studied in order to reduce the amount of wires in the **car**. ... Additional works will be carried out in order to find the best **configuration** for the modulation: ...
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... The communication speed is limited by the network bus **configuration** and no additional ... The standard **configuration** of a CAN bus node includes: a CAN transceiver, a CAN controller ... , the majority of **car** OEMs opts for this CAN-UDS tandem even in the latest high-end **car**'s ...
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E Kumaresyan, L Grozelda - 2014 6th International ..., 2014 - ieeexplore.ieee.org
... The longer transmission distance for **intra-car** sensor network can be obtained by **configuration** listed in Table 2. To provide the information of Bluetooth coverage, the measurement also conducted with the results are shown in Table 6. It is evident that the lowest received signal ...
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... As an example, Figure 6 shows the locations of the ZigBee and BLE transmitters and receivers in the sensor location **configuration** 1 (in ... Talty, and OK Tonguz, "Feasibility of safety

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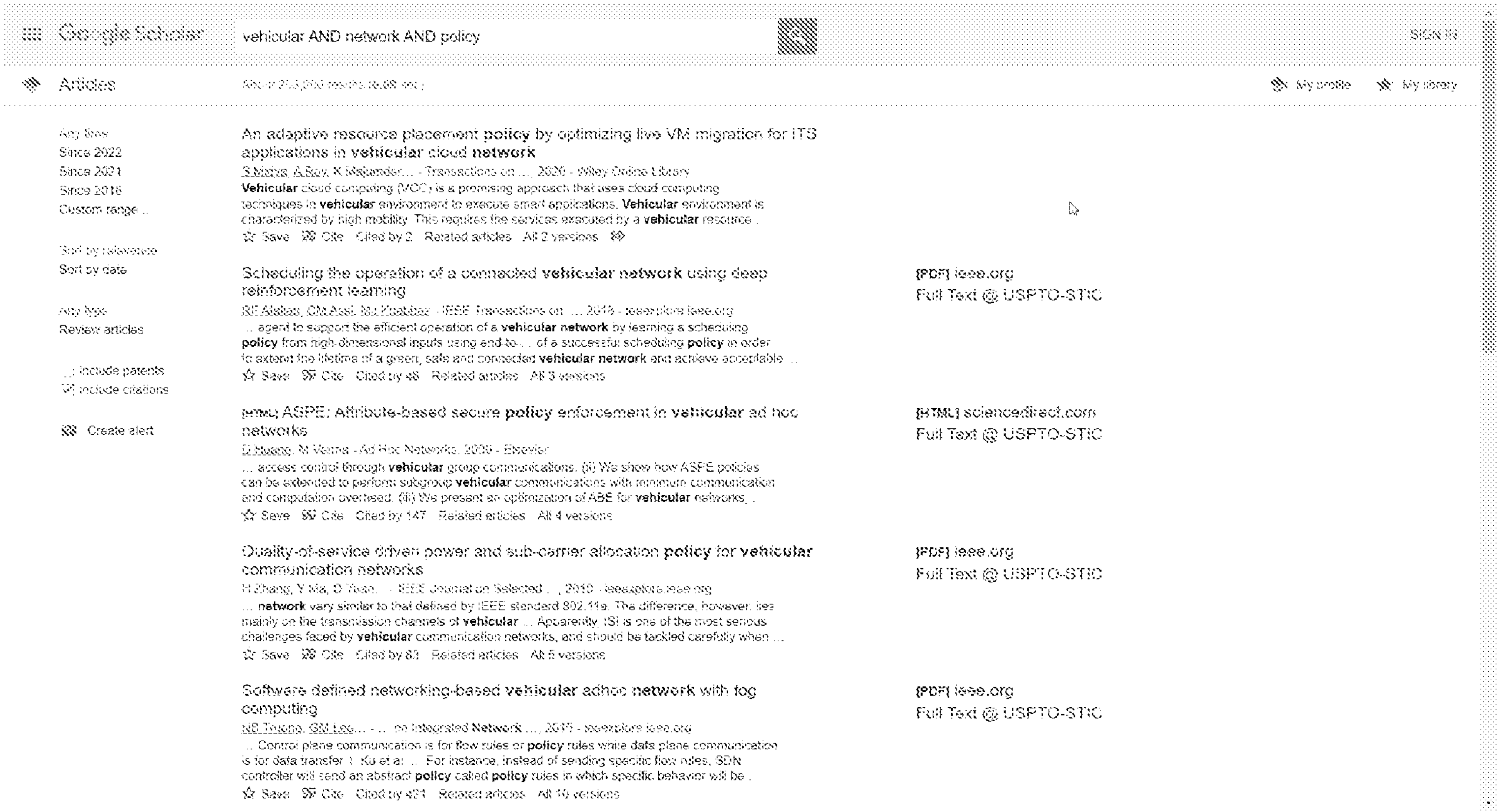

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H Zhang, Y Ma, D Yuan… - IEEE Journal on Selected …, 2010 - ieeexplore.ieee.org
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NB Truong, GM Lee… - … on Integrated **Network** …, 2015 - ieeexplore.ieee.org
… Control plane communication is for flow rules or **policy** rules while data plane communication is for data transfer [ ] Ku et al. … For instance, instead of sending specific flow rules, SDN controller will send an abstract **policy** called **policy** rules in which specific behavior will be …
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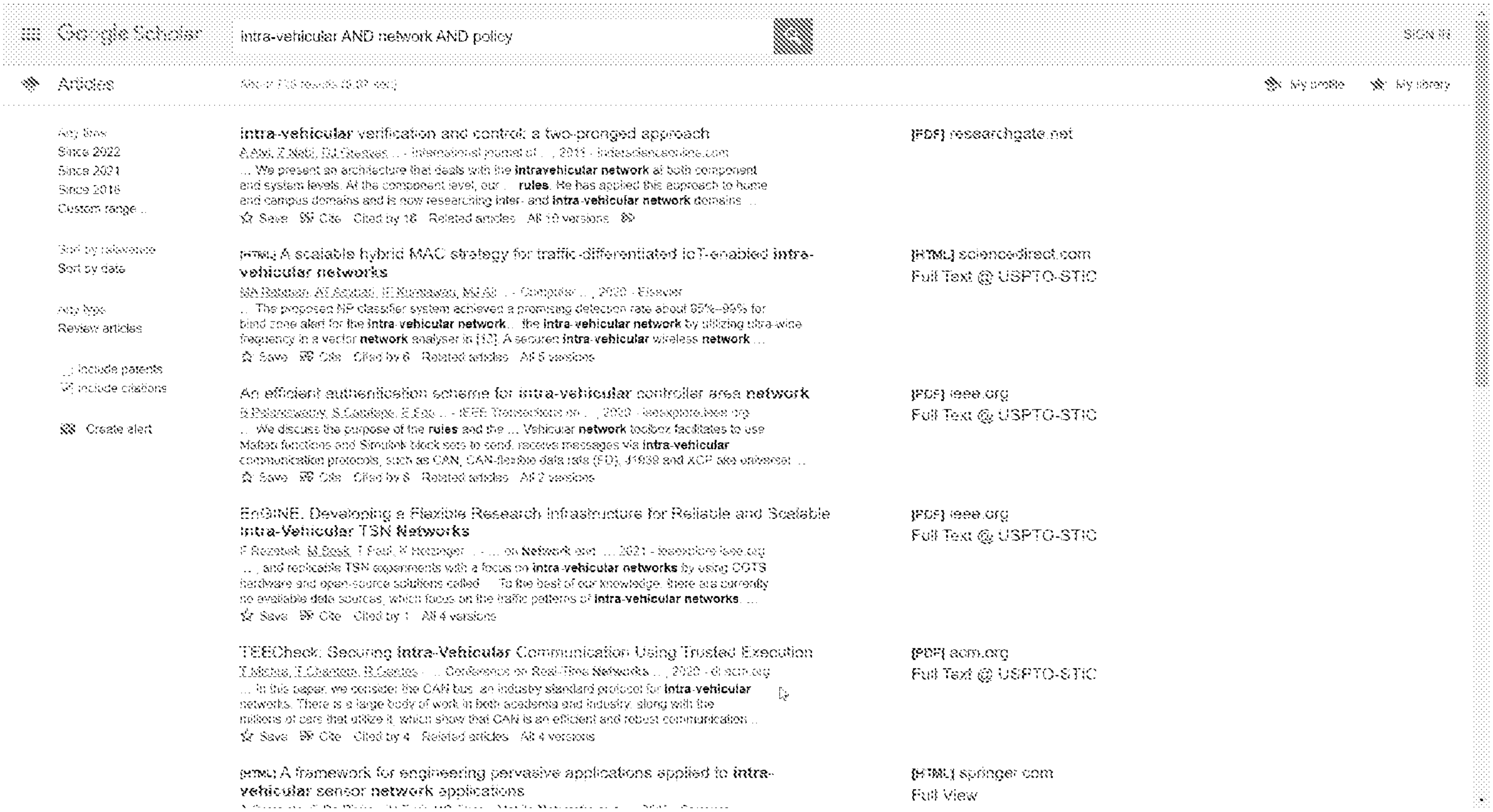

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A Alwi, Z Nabi, DJ Greaves ... - International journal of ..., 2011 - inderscienceonline.com
... We present an architecture that deals with the **intravehicular network** at both component and system levels. At the component level, our ... **rules**. He has applied this approach to home and campus domains and is now researching inter- and **intra-vehicular network** domains ...
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... The proposed NP classifier system achieved a promising detection rate about 95%–99% for blind zone alert for the **intra-vehicular network** ... the **intra-vehicular network** by utilizing ultra-wide frequency in a vector **network** analyser in [13]. A secured **intra-vehicular** wireless **network** ...
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B Palaniswamy, S Camtepe, E Foo ... - IEEE Transactions on ..., 2020 - ieeexplore.ieee.org
... We discuss the purpose of the **rules** and the ... Vehicular **network** toolbox facilitates to use Matlab functions and Simulink block sets to send, receive messages via **intra-vehicular** communication protocols, such as CAN, CAN-flexible data rate (FD), J1939 and XCP aka universal ...
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F Rezabek, M Bosk, T Paul, K Holzinger ... - ... on **Network** and ..., 2021 - ieeexplore.ieee.org
... , and replicable TSN experiments with a focus on **intra-vehicular networks** by using COTS hardware and open-source solutions called ... To the best of our knowledge, there are currently no available data sources, which focus on the traffic patterns of **intra-vehicular networks**. ...
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T Mishra, T Chantem, R Gerdes - ... Conference on Real-Time **Networks** ..., 2020 - dl.acm.org
... In this paper, we consider the CAN bus, an industry standard protocol for **intra-vehicular** networks. There is a large body of work in both academia and industry, along with the millions of cars that utilize it, which show that CAN is an efficient and robust communication ...
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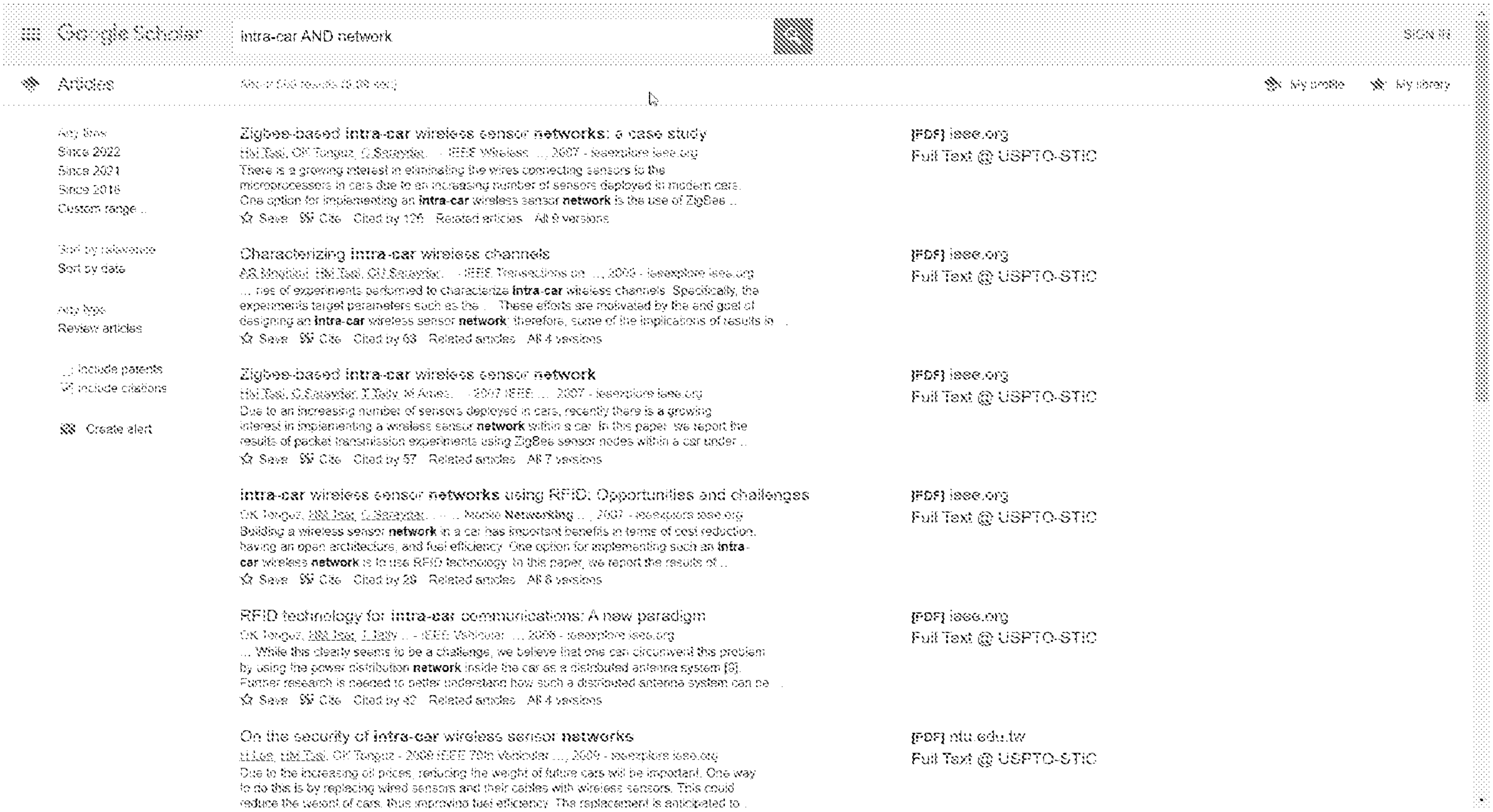

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Due to an increasing number of sensors deployed in cars, recently there is a growing interest in implementing a wireless sensor **network** within a car. In this paper, we report the results of packet transmission experiments using ZigBee sensor nodes within a car under ...
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Building a wireless sensor **network** in a car has important benefits in terms of cost reduction, having an open architecture, and fuel efficiency. One option for implementing such an **intra-car** wireless **network** is to use RFID technology. In this paper, we report the results of ...
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... While this clearly seems to be a challenge, we believe that one can circumvent this problem by using the power distribution **network** inside the car as a distributed antenna system [9]. Further research is needed to better understand how such a distributed antenna system can be ...
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Due to the increasing oil prices, reducing the weight of future cars will be important. One way to do this is by replacing wired sensors and their cables with wireless sensors. This could reduce the weight of cars, thus improving fuel efficiency. The replacement is anticipated to ...
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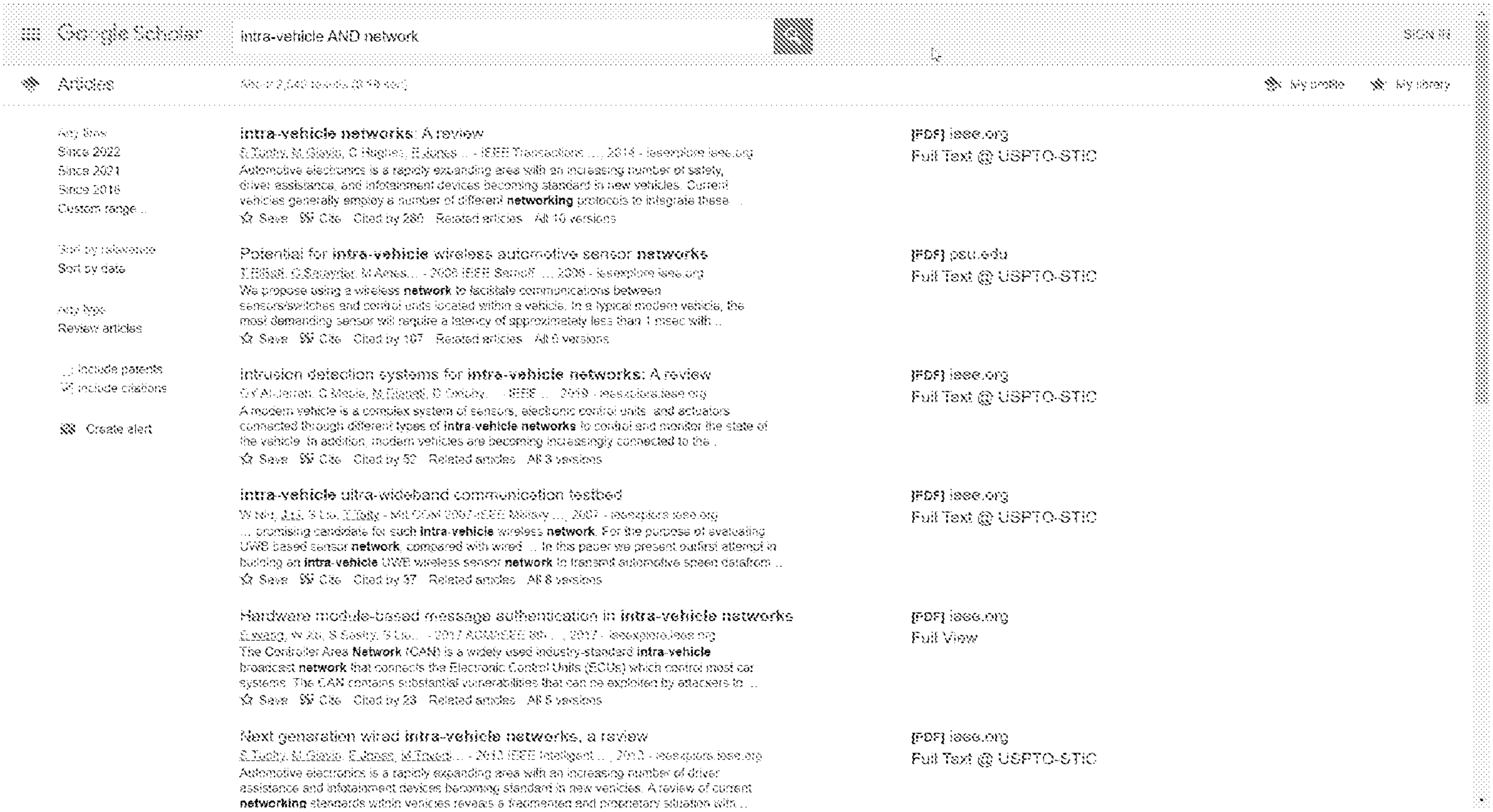

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S Tuohy, M Glavin, C Hughes, E Jones ... - IEEE Transactions ..., 2014 - ieeexplore.ieee.org

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We propose using a wireless **network** to facilitate communications between sensors/switches and control units located within a vehicle. In a typical modern vehicle, the most demanding sensor will require a latency of approximately less than 1 msec with ...

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OY Al-Jarrah, C Maple, M Dianati, D Oxtoby ... - IEEE ..., 2019 - ieeexplore.ieee.org

A modern vehicle is a complex system of sensors, electronic control units, and actuators connected through different types of **intra-vehicle networks** to control and monitor the state of the vehicle. In addition, modern vehicles are becoming increasingly connected to the ...

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W Niu, J Li, S Liu, T Talty - MILCOM 2007-IEEE Military ..., 2007 - ieeexplore.ieee.org

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S Tuohy, M Glavin, E Jones, M Trivedi ... - 2013 IEEE Intelligent ..., 2013 - ieeexplore.ieee.org

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